1, 3  √3  √3/  $\frac{}{3\sqrt{3}}$ → maximum

# Ch#14 - Multiple Integrals

## Ex#14.1 - Double Integrals

Double Integral = Volume Under the surface



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# 14.1

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1-12 Evaluate the iterated integrals.

1. $\int_0^1 \int_0^2 (x+3)\, dy\, dx$

Solve Inner Integral w.r.t y:

$$\int_0^2 (x+3)\, dy = [xy + 3y]_0^2$$

$$= (x(2) + 3(2)) - (x(0) + 3(0))$$

$$= 2x + 6$$

Outer Integral w.r.t x:

$$\int_0^1 (2x+6)\, dx = [x^2 + 6x]_0^1$$

$$= (1^2 + 6(1)) - (0)$$

$$= 7$$ Ans

2. $\int_1^3 \int_{-1}^1 (2x - 4y)\, dy\, dx$

Solve Inner Integral w.r.t y.

$$\int_{-1}^1 (2x - 4y)\, dy = [2xy - 2y^2]_{-1}^1$$

$$= (2x(1) - 2(1)^2) - (2x(-1) - 2(-1)^2)$$

$$= (2x - 2) - (-2x - 2)$$

$$= 2x - 2 + 2x + 2 = 4x$$

Outer Integral w.r.t x.



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$$\int_1^3 4x\,dx = [2x^2]_1^3 = (2(3)^2) - (2(1)^2) = 18-2$$

$$= 16 \quad \text{Ans}$$

3. $\int_2^4 \int_0^1 x^2 y\,dx\,dy$

Ans

Inner Integral w.r.t $x$.

$$\int_0^1 x^2 y\,dx = \left[\frac{x^3}{3} y\right]_0^1 = \left(\frac{1^3}{3} y\right) - (0) = \frac{1}{3} y$$

Outer Integral w.r.t $y$.

$$\int_2^4 \frac{1}{3} y\,dy = \left[\frac{y^2}{6}\right]_2^4 = \left(\frac{4^2}{6}\right) - \left(\frac{2^2}{6}\right) = \frac{16}{6} - \frac{4}{6} = \frac{12}{6}$$

$$= 2 \quad \text{Ans}$$

4. $\int_{-2}^0 \int_{-1}^2 (x^2+y^2)\,dx\,dy$

Ans

Inner Integral w.r.t $x$.

$$\int_{-1}^2 (x^2+y^2)\,dx = \left[\frac{x^3}{3} + xy^2\right]_{-1}^2$$

$$= \left(\frac{2^3}{3} + (2)y^2\right) - \left(\frac{(-1)^3}{3} + (-1)y^2\right)$$

$$= \left(\frac{8}{3} + 2y^2\right) - \left(-\frac{1}{3} - y^2\right) = \frac{9}{3} + 3y^2 = 3 + 3y^2$$

Outer Integral w.r.t $y$.

$$\int_{-2}^0 (3+3y^2)\,dy = [3y + y^3]_{-2}^0 = (0) - (3(-2) + (-2)^3)$$

$$= 0 - (-6 - 8) = 14 \quad \text{Ans}$$



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5. $\int_0^{\ln 3}\int_0^{\ln 2} e^{x+y}\,dy\,dx$

Inner Integral w.r.t y:

$$\int_0^{\ln 2}(e^x e^y)\,dy = e^x\int_0^{\ln x} e^y\,dy$$

$$= e^x\left[e^y\right]_0^{\ln 2}$$

$$= e^x(e^{\ln 2} - e^0)$$

$$= e^x(2-1) = e^x$$

Outer Integral w.r.t x:

$$\int_0^{\ln 3} e^x\,dx = \left[e^x\right]_0^{\ln 3} = e^{\ln 3} - e^0 = 3 - 1 = 2$$ Ans

6. $\int_0^2\int_0^1 y\sin x\,dy\,dx$

Inner Integral w.r.t y:

$$\int_0^1 (y\sin x)\,dy = \sin x\int_0^1 y\,dy$$

$$= \sin x\left[\frac{y^2}{2}\right]_0^1 = \sin x\left(\frac{1^2}{2} - 0\right)$$

$$= \frac{1}{2}\sin x$$

Outer Integral w.r.t x:

$$\int_0^2 \frac{1}{2}\sin x\,dx = \left[-\frac{1}{2}\cos x\right]_0^2 = \left(-\frac{1}{2}\cos(2)\right) - \left(-\frac{1}{2}\cos(0)\right)$$

$$= -\frac{1}{2}\cos(2) + \frac{1}{2} = \frac{1-\cos 2}{2}$$ Ans



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7. $\int_{-1}^{0}\int_{2}^{5} dx\,dy$

Inner Integral w.r.t $x$:

$$\int_{2}^{5} 1\,dx = [x]_{2}^{5} = 5-2 = 3$$

Outer Integral w.r.t $y$:

$$\int_{-1}^{0} 3\,dy = [3y]_{-1}^{0} = (3(0)) - (3(-1))$$
$$= 0 + 3 = 3$$ ans

8. $\int_{4}^{6}\int_{-3}^{7} dy\,dx$

Inner Integral w.r.t $y$:

$$\int_{-3}^{7} 1\,dy = [y]_{-3}^{7} = 7-(-3) = 10$$

Outer Integral w.r.t $x$

$$\int_{4}^{6} 10\,dx = [10x]_{4}^{6} = 10(6) - 10(4)$$
$$= 60 - 40 = 20$$ ans

9. $\int_{0}^{1}\int_{0}^{1} \frac{x}{(xy+1)^2}\,dy\,dx$

Inner Integral w.r.t $y$

$$\int_{0}^{1} \frac{x}{(xy+1)^2}\,dy$$

Let $u = xy + 1$ , $du = x\,dy$



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$$\int_0^1 u^{-2}\,du = \left[-\frac{1}{u}\right]_0^1 = \left[-\frac{1}{xy+1}\right]_0^1$$

$$= \left(-\frac{1}{x+1}\right) - \left(-\frac{1}{0+1}\right) = 1 - \frac{1}{x+1}$$

Outer Integral w.r.t $x$:

$$\int_0^1 \left(1 - \frac{1}{x+1}\right) dx = \left[x - \ln|x+1|\right]_0^1$$

$$= (1 - \ln(2)) - (0 - \ln(1))$$

$$= 1 - \ln 2 \quad \text{Ans}$$

10. $$\int_{\frac{\pi}{2}}^{\pi} \int_1^2 x \cos xy \, dy \, dx$$

Inner Integral w.r.t $y$:

$$\int_1^2 x \cos xy \, dy$$

Let $u = xy$, $du = x\,dy$

$$\int_1^2 \cos(u)\,du = [\sin(u)]_1^2 = [\sin(xy)]_1^2$$

$$= (\sin(2x)) - (\sin(x)) = \sin 2x - \sin x$$

Outer Integral w.r.t $x$:

$$\int_{\frac{\pi}{2}}^{\pi} (\sin(2x) - \sin(x))\,dx = \left[-\frac{1}{2}\cos(2x) + \cos(x)\right]_{\pi/2}^{\pi}$$

$$= \left(-\frac{1}{2}\cos(2\pi) + \cos(\pi)\right) - \left(-\frac{1}{2}\cos(\pi) + \cos\left(\frac{\pi}{2}\right)\right)$$

$$= \left(-\frac{1}{2}(1) + (-1)\right) - \left(-\frac{1}{2}(-1) + 0\right) = -\frac{3}{2} - \frac{1}{2} = -2 \quad \text{Ans}$$



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11. $\int_0^{\ln 2}\int_0^1 xye^{y^2x}\,dy\,dx$

Solution: Inner Integral w.r.t $y$

$$\int_0^1 xye^{y^2x}\,dy$$

Let $u = y^2x$, $du = 2yx\,dy \Rightarrow xy\,dy = \frac{1}{2}du$

$$\int_0^1 \frac{1}{2}e^u\,du = \left[\frac{1}{2}e^u\right]_0^1 = \left[\frac{1}{2}e^{y^2x}\right]_0^1$$

$$= \left(\frac{1}{2}e^{(1)^2x}\right) - \left(\frac{1}{2}e^{(0)^2x}\right) = \frac{1}{2}e^x - \frac{1}{2}e^0$$

$$= \frac{1}{2}(e^x - 1)$$

Outer Integral w.r.t $x$:

$$\int_0^{\ln 2}\frac{1}{2}(e^x - 1)\,dx = \left[\frac{1}{2}(e^x - x)\right]_0^{\ln 2}$$

$$= \left(\frac{1}{2}(e^{\ln 2} - \ln 2)\right) - \left(\frac{1}{2}(e^0 - 0)\right)$$

$$= \left(\frac{1}{2}(2 - \ln 2)\right) - \left(\frac{1}{2}(1)\right)$$

$$= 1 - \frac{1}{2}\ln 2 - \frac{1}{2}$$

$$= -\frac{1}{2}\ln 2 + \frac{1}{2}$$

$$= \frac{1 - \ln 2}{2}$$ Ans



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# 14.1

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12. $\int_3^4 \int_1^2 \frac{1}{(x+y)^2}\, dy\, dx$

Solve

Inner Integral w.r.t $y$:

$$\int_1^2 (x+y)^{-2}\, dy = \left[-\frac{1}{x+y}\right]_0^2 = \left(-\frac{1}{x+2}\right) - \left(-\frac{1}{x+1}\right)$$

$$= \frac{1}{x+1} - \frac{1}{x+2}$$

Outer Integral w.r.t $x$:

$$\int_3^4 \left(\frac{1}{x+1} - \frac{1}{x+2}\right) dx = \left[\ln|x+1| - \ln|x+2|\right]_3^4$$

$$= \left[\ln\left(\frac{x+1}{x+2}\right)\right]_3^4$$

$$= \ln\left(\frac{5}{6}\right) - \ln\left(\frac{4}{5}\right) = \ln\left(\frac{5/6}{4/5}\right)$$

$$= \ln\left(\frac{25}{24}\right)$$ Ans

Evaluate the double integral over the rectangular region R.

13. $\iint_R 4xy^3\, dA;\ R = \{(x,y) : -1 \le x \le 1,\ -2 \le y \le 2\}$

Solve

$$\int_{-1}^1 \int_{-2}^2 4xy^3\, dy\, dx$$

Inner Integral w.r.t $y$:

$$\int_{-2}^2 4xy^3\, dy = 4x\left[\frac{y^4}{4}\right]_{-2}^2$$



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$$= 4x\left(\frac{2^4}{4} - \frac{(-2)^4}{4}\right) = 4x\left(\frac{16}{4} - \frac{16}{4}\right)$$

$$= 4x(0) = 0$$

Outer Integral w.r.t $x$.

$$\int_{-1}^{1} 0\,dx = 0$$ ans

# Ex # 14.2

1-8 Evaluate the iterated integral.

1. $\int_0^1 \int_{x^2}^{x} xy^2\,dy\,dx$

Inner Integral w.r.t $y$:

$$\int_{x^2}^{x} xy^2\,dy = x\left[\frac{y^3}{3}\right]_{x^2}^{x} = x\left(\frac{(x)^3}{3} - \frac{(x^2)^3}{3}\right)$$

$$= x\left(\frac{x^3}{3} - \frac{x^6}{3}\right) = \frac{1}{3}x^4 - \frac{1}{3}x^7.$$

Outer Integral w.r.t $x$.

$$\int_0^1 \left(\frac{1}{3}x^4 - \frac{1}{3}x^7\right)dx = \left[\frac{1}{3}\cdot\frac{x^5}{5} - \frac{1}{3}\cdot\frac{x^8}{8}\right]_0^1$$

$$= \left(\frac{1}{15} - \frac{1}{24}\right) - (0) = \frac{8-5}{120}$$

$$= \frac{3}{120} = \frac{1}{40}$$ ans



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# 14.2

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2. $\int_1^{3/2}\int_y^{3-y} y\,dx\,dy$

Inner Integral w.r.t $x$:

$$\int_y^{3-y} y\,dx = y[x]_y^{3-y} = y((3-y)-(y)) = y(3-2y)$$

$$= 3y - 2y^2$$

Outer Integral w.r.t $y$:

$$\int_1^{3/2}(3y-2y^2)dy = \left[\frac{3y^2}{2} - \frac{2y^3}{3}\right]_1^{3/2}$$

$$= \left(\frac{3(3/2)^2}{2} - \frac{2(3/2)^3}{3}\right) - \left(\frac{3(1)^2}{2} - \frac{2(1)^3}{3}\right)$$

$$= \left(\frac{27}{8} - \frac{54}{24}\right) - \left(\frac{3}{2} - \frac{2}{3}\right) = \left(\frac{9}{8}\right) - \left(\frac{5}{6}\right)$$

$$= \frac{27-20}{24} = \frac{7}{24}$$ Ans

3. $\int_0^3\int_0^{\sqrt{9-y^2}} y\,dx\,dy$

Inner Integral w.r.t $x$:

$$\int_0^{\sqrt{9-y^2}} y\,dx = y[x]_0^{\sqrt{9-y^2}} = y(\sqrt{9-y^2} - 0)$$

$$= y\sqrt{9-y^2}$$

Outer Integral w.r.t $y$:

$$\int_0^3 y\sqrt{9-y^2}\,dy$$



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Let $u = 9 - y^2$, $du = -2y\,dy$

$y\,dy = -\frac{1}{2}\,du$

$$\int_0^3 \sqrt{u}\left(-\frac{1}{2}\,du\right) = \left[-\frac{1}{2}\,\frac{u^{3/2}}{3/2}\right]_0^3 = \left[-\frac{1}{3}(9-y^2)^{3/2}\right]_0^3$$

$$= \left(-\frac{1}{3}(9-3^2)^{3/2}\right) - \left(-\frac{1}{3}(9-0^2)^{3/2}\right)$$

$$= \left(-\frac{1}{3}(0)^{3/2}\right) - \left(-\frac{1}{3}(9)^{3/2}\right) = 0 - \left(-\frac{1}{3}(27)\right)$$

$$= 9$$ Ans

5. $$\int_{\sqrt{\pi}}^{\sqrt{2\pi}} \int_0^{x^3} \sin\left(\frac{y}{x}\right) dy\,dx.$$

Sol: Inner Integral w.r.t $y$:

$$\int_0^{x^3} \sin\left(\frac{y}{x}\right) dy$$

Let $u = y/x$. Then $du = (1/x)\,dy$

$dy = x\,du$

$$\int_0^{x^3} \sin(u)\,(x\,du) = x\int_0^{x^3} \sin(u)\,du$$

$$= \left[-x\cos(u)\right]_0^{x^3}$$

$$= \left[-x\cos(y/x)\right]_0^{x^3}$$

$$= (-x\cos(x^3/x)) - (-x\cos(0))$$

$$= -x\cos(x^2) - (-x\cdot 1)$$

$$= x - x\cos(x^2).$$



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# 14.2

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Outer Integral w.r.t x:

$$\int_{\sqrt{\pi}}^{\sqrt{2\pi}} (x - x\cos(x^2))\,dx = \left[\frac{x^2}{2} - \frac{1}{2}\sin(x^2)\right]_{\sqrt{\pi}}^{\sqrt{2\pi}}$$

$$= \left(\frac{(\sqrt{2\pi})^2}{2} - \frac{1}{2}\sin(2\pi)\right) - \left(\frac{(\sqrt{\pi})^2}{2} - \frac{1}{2}\sin(\pi)\right)$$

$$= \left(\frac{2\pi - 0}{2}\right) - \left(\frac{\pi}{2} - 0\right) = \pi - \frac{\pi}{2} = \frac{\pi}{2}$$ Ans

4. $$\int_{1/4}^{1}\int_{x^2}^{x}\sqrt{\frac{x}{y}}\,dy\,dx$$

Solut Inner Integral w.r.t y:

$$\int_{x^2}^{x} x^{1/2} y^{-\frac{1}{2}}\,dy = x^{1/2}\left[\frac{y^{1/2}}{1/2}\right]_{x^2}^{x}$$

$$= 2x^{1/2}\left((x)^{1/2} - (x^2)^{1/2}\right)$$

$$= 2x^{1/2}\left(x^{\frac{1}{2}} - x\right) = 2x - 2x^{3/2}$$

Outer Integral w.r.t x:

$$\int_{1/4}^{1} (2x - 2x^{3/2})\,dx = \left[x^2 - \frac{2x^{5/2}}{5/2}\right]_{1/4}^{1}$$

$$= \left[x^2 - \frac{2x^{5/2}}{5/2}\right]_{1/4}^{1} = \left[x^2 - \frac{4}{5}x^{5/2}\right]_{1/4}^{1}$$

$$= \left(1 - \frac{4}{5}\right) - \left(\left(\frac{1}{4}\right)^2 - \frac{4}{5}\left(\frac{1}{4}\right)^{5/2}\right)$$

$$= \left(\frac{1}{5}\right) - \left(\frac{1}{16} - \frac{4}{5}\cdot\frac{1}{32}\right) = \frac{1}{5} - \left(\frac{1}{16} - \frac{1}{40}\right)$$

$$= \frac{1}{5} - \frac{3}{80} = \frac{13}{80}$$ Ans



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# 14.2

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6. $\int_{-1}^{1}\int_{-x^2}^{x^2}(x^2-y)\,dy\,dx$

Inner Integral w.r.t y:

$$\int_{-x^2}^{x^2}(x^2-y)\,dy=\left[x^2y-\frac{y^2}{2}\right]_{-x^2}^{x^2}$$

$$=\left(x^2(x^2)-\frac{(x^2)^2}{2}\right)-\left(x^2(-x^2)-\frac{(-x^2)^2}{2}\right)$$

$$=\left(x^4-\frac{x^4}{2}\right)-\left(-x^4-\frac{x^4}{2}\right)=\left(\frac{1}{2}x^4\right)-\left(-\frac{3}{2}x^4\right)$$

$$=2x^4$$

Outer Integral w.r.t x:

$$\int_{-1}^{-1}2x^4\,dx=\left[\frac{2x^5}{5}\right]_{-1}^{1}=\left(\frac{2(1)^5}{5}-\left(\frac{2(-1)^5}{5}\right)\right.$$

$$=\frac{2}{5}-\left(-\frac{2}{5}\right)=\frac{4}{5}$$

7. $\int_0^1\int_0^x y\sqrt{x^2-y^2}\,dy\,dx$

Inner Integral w.r.t y:

$$\int_0^x y\sqrt{x^2-y^2}\,dy$$

Let $u=x^2-y^2$, $du=-2y\,dy$

$$y\,dy=-\frac{1}{2}du$$

$$\int_0^x\sqrt{u}\left(-\frac{1}{2}\,du\right)=-\frac{1}{2}\cdot\frac{u^{3/2}}{3/2}=-\frac{1}{3}u^{3/2}$$

$$=\left[-\frac{1}{3}\left(x^2-y^2\right)^{3/2}\right]_0^x$$



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$$= \left(-\frac{1}{3}(x^2 - x^2)^{3/2}\right) - \left(-\frac{1}{3}(x^2 - 0^2)^{3/2}\right)$$

$$= 0 - \left(-\frac{1}{3}(x^2)^{3/2}\right) = -\left(-\frac{1}{3}x^3\right) = \frac{1}{3}x^3$$

Outer Integral w.r.t $x$:

$$\int_0^1 \frac{1}{3}x^3\,dx = \left[\frac{1}{3}\cdot\frac{x^4}{4}\right]_0^1 = \left[\frac{x^4}{12}\right]_0^1$$

$$= \frac{1}{12} - 0 = \frac{1}{12}$$ Ans

8. $\int_1^2 \int_0^{y^2} e^{x/y^2}\,dx\,dy$

Inner Integral w.r.t $x$:

$$\int_0^{y^2} e^{x/y^2}\,dx$$

Let $u = x/y^2$, $du = (1/y^2)\,dx$

$dx = y^2\,du$

$$\int_0^{y^2} e^u (y^2\,du) = y^2 \int_0^{y^2} e^u\,du$$

$$= \left[y^2 e^u\right]_0^{y^2} = \left[y^2 e^{x/y^2}\right]_0^{y^2} = (y^2 e^{x/y^2}) - (y^2 e^{0/y^2})$$

$$= (y^2 e^1) - (y^2 e^0) = y^2 e - y^2(1) = y^2(e-1)$$

Outer Integral w.r.t $y$:

$$\int_1^2 y^2(e-1)\,dy = (e-1)\int_1^2 y^2\,dy = (e-1)\left[\frac{y^3}{3}\right]_1^2$$

$$= (e-1)\left(\frac{2^3}{3} - \frac{1^3}{3}\right) = (e-1)\left(\frac{8-1}{3}\right) = \frac{7(e-1)}{3}$$ Ans



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Maxima at (-1, 5), no minima.
Reason: Function is 1 - (sum of squares), which is largest when squares are zero



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# Ex # 14.5

Date

1-6 Evaluate the iterated Integral.

1. $\int_{-1}^{1}\int_{0}^{2}\int_{0}^{1}(x+y+z^2)\,dx\,dy\,dz$.

Solve

Innermost Integral w.r.t $x$:

$$\int_0^1 (x^2+y^2+z^2)\,dx = \left[\frac{x^3}{3}+xy^2+xz^2\right]_0^1$$

$$= \left(\frac{1}{3}+y^2+z^2\right)-0 = \frac{1}{3}+y^2+z^2$$

Middle Integral w.r.t $y$:

$$\int_0^2 \left(\frac{1}{3}+y^2+z^2\right)dy = \left[\frac{1}{3}y+\frac{y^3}{3}+yz^2\right]_0^2$$

$$= \left(\frac{2}{3}+\frac{8}{3}+2z^2\right)-(0) = \frac{10}{3}+2z^2$$

Outer Integral w.r.t $z$:

$$\int_{-1}^{1}\left(\frac{10}{3}+2z^2\right)dz = \left[\frac{10z}{3}+\frac{2z^3}{3}\right]_{-1}^{1}$$

$$= \left(\frac{10}{3}+\frac{2}{3}\right)-\left(-\frac{10}{3}-\frac{3}{2}\right) = \left(\frac{12}{3}\right)-\left(-\frac{12}{3}\right)$$

$$= 4-(-4) = 8$$ Ans

2. $\int_{1/3}^{1/2}\int_{0}^{\pi}\int_{0}^{1} zx\sin xy\,dz\,dy\,dx$

Solve

Innermost Integral w.r.t $z$:

$$\int_0^1 zx\sin(xy)\,dz = x\sin(xy)\left[\frac{z^2}{2}\right]_0^1$$

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$$= x\sin(xy)\left(\frac{1}{2}-0\right) = \frac{1}{2}x\sin(xy)$$

Middle Integral w.r.t y:

$$\int_0^{\pi} \frac{1}{2}x\sin(xy)\,dy = \left[\frac{1}{2}\left(-\cos(xy)\right)\right]_0^{\pi}$$

$$= \left[-\frac{1}{2}\cos(xy)\right]_0^{\pi}$$

$$= \left(-\frac{1}{2}\cos(\pi x)\right) - \left(-\frac{1}{2}\cos(0)\right) = \frac{1}{2}\left(1-\cos(\pi x)\right)$$

Outer Integral w.r.t x:

$$\int_{1/3}^{1/2} \frac{1}{2}\left(1-\cos(\pi x)\right)dx = \frac{1}{2}\left[x - \frac{\sin(\pi x)}{\pi}\right]_{1/3}^{1/2}$$

$$= \frac{1}{2}\left[\left(\frac{1}{2} - \frac{\sin(\pi/2)}{\pi}\right) - \left(\frac{1}{3} - \frac{\sin(\pi/3)}{\pi}\right)\right]$$

$$= \frac{1}{2}\left[\left(\frac{1}{2} - \frac{1}{\pi}\right) - \left(\frac{1}{3} - \frac{\sqrt{3}}{2\pi}\right)\right]$$

$$= \frac{1}{2}\left[\frac{1}{6} - \frac{1}{\pi} + \frac{\sqrt{3}}{2\pi}\right] = \frac{1}{12} + \frac{\sqrt{3}-2}{4\pi}$$

3. $\int_0^2 \int_{-1}^{y^2} \int_{-1}^{z} yz\,dx\,dz\,dy$

Innermost Integral w.r.t x:

$$\int_{-1}^{z} yz\,dx = yz\left[x\right]_{-1}^{z} = yz\left(z-(-1)\right)$$

$$= yz(z+1) = yz^2 + yz$$

Middle Integral w.r.t z:

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$$\int_{-1}^{y^2} (yz^2 + yz)\, dz = y\left[\frac{z^3}{3} + \frac{z^2}{2}\right]_{-1}^{y^2}$$

$$= y\left[\left(\frac{(y)^{2\cdot 3}}{3} + \frac{(y^2)^2}{2}\right) - \left(\frac{(-1)^3}{3} + \frac{(-1)^2}{2}\right)\right]$$

$$= y\left(\left(\frac{y^6}{3} + \frac{y^4}{2}\right) - \left(-\frac{1}{3} + \frac{1}{2}\right)\right)$$

$$= y\left(\frac{y^6}{3} + \frac{y^4}{2} - \frac{1}{6}\right) = \frac{y^7}{3} + \frac{y^5}{2} - \frac{y}{6}$$

Outer Integral w.r.t y:

$$\int_0^2 \left(\frac{y^7}{3} + \frac{y^5}{2} - \frac{y}{6}\right) dy = \left[\frac{y^8}{24} + \frac{y^6}{12} - \frac{y^2}{12}\right]_0^2$$

$$= \left(\frac{2^8}{24} + \frac{2^6}{12} - \frac{2^2}{12}\right) = \frac{32}{3} + \frac{16}{3} - \frac{1}{3} = \frac{47}{3}$$ Ans

4. $\int_0^{\pi/4} \int_0^1 \int_0^{x^2} x \cos y \, dz\, dx\, dy$

Innermost Integral w.r.t z:

$$\int_0^{x^2} x \cos y \, dz = x\cos(y)\left[z\right]_0^{x^2} = x\cos y\,(x^2 - 0)$$

$$= x^3 \cos y$$

Middle Integral w.r.t x:

$$\int_0^1 x^3 \cos y \, dx = \cos y\left[\frac{x^4}{4}\right]_0^1 = \cos y\left(\frac{1}{4} - 0\right)$$

$$= \frac{1}{4}\cos(y)$$

Outermost Integral w.r.t y:

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Date

$$\int_0^{\pi/4} \frac{1}{4}\cos(y)\,dy = \left[\frac{1}{4}\sin(y)\right]_0^{\pi/4}$$

$$= \frac{1}{4}\sin\left(\frac{\pi}{4}\right) - \frac{1}{4}\sin(0) = \frac{1}{4}\left(\frac{\sqrt{2}}{2}\right) - 0$$

$$= \frac{\sqrt{2}}{8} \quad \text{Ans}$$

5. $\int_0^3 \int_0^{\sqrt{9-z^2}} \int_0^x xy\,dy\,dx\,dz$

Innermost Integral w.r.t y:

$$\int_0^x xy\,dy = x\left[\frac{y^2}{2}\right]_0^x = x\left(\frac{x^2}{2} - 0\right) = \frac{1}{2}x^3$$

Middle Integral w.r.t x:

$$\int_0^{\sqrt{9-z^2}} \frac{1}{2}x^3\,dx = \frac{1}{2}\left[\frac{x^4}{4}\right]_0^{\sqrt{9-z^2}} = \frac{1}{8}\left[x^4\right]_0^{\sqrt{9-z^2}}$$

$$= \frac{1}{8}\left(\left(\sqrt{9-z^2}\right)^4 - 0\right) = \frac{1}{8}(9-z^2)^2$$

$$= \frac{1}{8}(81 - 18z^2 + z^4)$$

Outer Integral w.r.t z:

$$\int_0^3 \frac{1}{8}(81 - 18z^2 + z^4)\,dz = \frac{1}{8}\left[81z - \frac{18z^3}{3} + \frac{z^5}{5}\right]_0^3$$

$$= \frac{1}{8}\left[81z - 6z^3 + \frac{z^5}{5}\right]_0^3$$

$$= \frac{1}{8}\left((81)(3) - 6(3)^3 + \frac{3^5}{5}\right) - 0\Big)$$

$$= \frac{1}{8}\left(243 - 162 + \frac{243}{5}\right) = \frac{1}{8}\left(81 + \frac{243}{5}\right)$$

$$= \frac{81}{5} \quad \text{Ans}$$

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Date

6. $\int_1^3 \int_x^{x^2} \int_0^{\ln z} x e^y \, dy \, dz \, dx$

solve

Innermost Integral w.r.t y:

$$\int_0^{\ln z} x e^y \, dy = x \left[ e^y \right]_0^{\ln z} = x(e^{\ln z} - e^0)$$

$$= x(z - 1)$$

Middle Integral w.r.t z:

$$\int_x^{x^2} x(z-1) \, dz = x \left[ \frac{z^2}{2} - z \right]_x^{x^2}$$

$$= x \left[ \left( \frac{(x^2)^2}{2} - x^2 \right) - \left( \frac{x^2}{2} - x \right) \right]$$

$$= x \left( \frac{x^4}{2} - x^2 - \frac{x^2}{2} + x \right) = x \left( \frac{x^4}{2} - \frac{3x^2}{2} + x \right)$$

$$= \frac{x^5}{2} - \frac{3x^3}{2} + x^2$$

Outermost Integral w.r.t x:

$$\int_1^3 \left( \frac{x^5}{2} - \frac{3x^3}{2} + x^2 \right) dx = \left[ \frac{x^6}{12} - \frac{3x^4}{8} + \frac{x^3}{3} \right]_1^3$$

$$= \left( \frac{3^6}{12} - \frac{3(3^4)}{8} + \frac{3^3}{3} \right) - \left( \frac{1}{12} - \frac{3}{8} + \frac{1}{3} \right)$$

$$= \left( \frac{729}{12} - \frac{243}{8} + 9 \right) - \left( \frac{2 - 9 + 8}{24} \right)$$

$$= \left( \frac{243}{4} - \frac{243}{8} + 9 \right) - \left( \frac{1}{24} \right)$$

$$= \left( \frac{243}{8} + \frac{72}{8} \right) - \frac{1}{24} = \frac{315}{8} - \frac{1}{24} = \frac{944}{24} = \frac{118}{3}$$ Ans.